# Bolted Joint — bench sheet

M6 (Ø6 mm · As 20.1 mm²) · class 8.8 (medium-carbon, Q&T) · Dry steel, plain (K ≈ 0.20) · no washers

Plate 1 Aluminum 6061-T6 × 8.00 mm · plate 2 Mild steel (S235) × 12.00 mm · grip 20.00 mm · preload target 65% of proof — conservative (default)

T = 6.00 N·m → Fi 5.00 kN · external load P = 500 N · MechCalc — design check, not a substitute for full analysis

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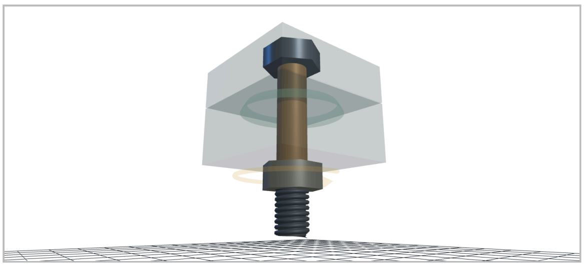
**n = 1.80** σred 322 MPa vs proof 580 MPa while torquing · clamp left 4.63 kN at P · separation n 13.49



The joint as modelled at T = 6.00 N·m: the cone is the clamp load spreading through the plates, shaded by local pressure; bolt and plates are coloured by how close each is to its own limit. Deflections exaggerated ×40 so the motion is visible.

## Joint at a glance

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|---|---|
| Preload Fi from T | **5.00 kN** |
| Reduced σred (vM) while torquing | **322 MPa · n 1.80 vs proof** |
| Working σ after torsion relaxes | **255 MPa · n 2.51 vs yield** |
| Stiffness kb / km · load share C | **201 / 576 kN/mm · C 0.259** |
| Bolt force / clamp left at P | **5.13 kN / 4.63 kN** |
| Separation load Psep | **6.74 kN · n 13.49** |
| Bearing p under head/nut | **161 MPa · n 1.86 / 3.04** |
| Interface pressure · cone Ø | **20 MPa · Ø18.2 mm** |
| Bolt stretch / plates squash | **24.9 / 8.7 µm** |
| Recommended torque | **9.09 N·m · limited by the bolt** |

## Equations used

$F = T / (K \cdot d)$ — Preload from tightening torque — nut-factor form

$\sigma = F / A_s$ — Direct tensile stress on the stress area As

$\tau = 16 \cdot T_{th} / (\pi \cdot d_s^3)$, $T_{th} \approx 0.5 \cdot T$ — Torsion from thread friction while torquing

$\sigma_{red} = \sqrt{\sigma^2 + 3\tau^2}$ — Reduced (von Mises) stress during tightening — VDI 2230

$n = S_p / \sigma_{red}$ — Safety factor against proof strength

$k_b = E \cdot A_s / L$ — Bolt as a tension spring over the grip

$k_m = f(30° \text{ cone frusta})$ — Member stiffness — Shigley pressure-cone stack

$C = k_b / (k_b + k_m)$ — Stiffness ratio: bolt's share of external load

$F_b = F_i + C \cdot P$, $F_m = F_i - (1-C) \cdot P$ — Load sharing: bolt force & remaining clamp

$P_{sep} = F_i / (1 - C)$ — External load at which the joint separates

$p = F / [\pi(d_w^2 - d_h^2)/4]$ — Bearing pressure under head/nut vs plate limit pG

$T_{rec} = K \cdot d \cdot \min(0.65 \cdot S_p \cdot A_s, 0.9 \cdot p_G \cdot A_{bear})$ — Recommended torque — bolt or plate, whichever gives out first

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Closed-form design check, not FEA. Nut-factor torque model (K scatters ±25% between real joints — lubricate for consistency), fully-threaded fastener for kb, Shigley 30° cone frusta for km, concentric and purely tensile external load. Typical reference values — verify before production use.